set. The logic is Boolean, and the evolution of states over time is deterministic and reversible. In the simplest examples we considered, the state-space consisted of a few points: Heads and Tails for a coin, {1, 2, 3, 4, 5, 6} for a die. The states were pictured as a set of points on the page, and the time evolution was just a rule telling you where to go next. A law of motion consisted of a graph with arrows connecting the states. The main rule—determinism—was that wherever you are in the state-space, the next state is completely specified by the law of motion. But there was also another rule called reversibility. Reversibility is the requirement that a properly formulated law must also tell you where you were last. A good law corresponds to a graph with exactly one arrow in and one arrow out at each state.

There is another way to describe these requirements. I called it the *minus first law*, because it underlies everything else. It says that information is never lost. If two identical isolated systems start out in different states, they stay in different states. Moreover, in the past they were also in different states. On the other hand, if two identical systems are in the same state at some point in time, then their histories and their future evolutions must also be identical. Distinctions are conserved. The quantum version of the minus first law has a name—*unitarity*.

4.2 UNITARITY

Let us consider a closed system that at time t is in the quantum state $|\Psi\rangle$. (The use of the Greek letter Ψ [psi] for quantum states is traditional when considering the evolution of systems.) To indicate that the state was $|\Psi\rangle$ at the specific time t, let's complicate the notation a bit and call the state $|\Psi(t)\rangle$. Of course, this notation suggests a bit more than just "the state was $|\Psi\rangle$ at time t." It also suggests that the state may be different at different times. Thus, we think of $|\Psi(t)\rangle$ as representing the entire history of the system.

The basic dynamical assumption of quantum mechanics is that if you know the state at one time, then the quantum equations of motion tell you what it will be later. Without loss of generality, we can take the initial time to